

QUIPUS B1

Physical PCB placement update

Addendum to the schematic and assembly guide; read with the current B1 CAD package.

PLACEMENT UPDATED. COPPER ROUTING IS NOT COMPLETE.

The B1 carrier has no copper tracks or standalone routed vias. Native ERC/DRC, detailed routing and manufacturing-output review are still required.

Board envelope	Electrical content	Current routing
65 x 125 x 1.6 mm Four copper layers	180 references 113 nets	0 copper tracks 0 routed vias

Why the board is longer

The earlier 65 x 105 mm outline becomes 65 x 125 mm. The added 20 mm reserves a 41 x 29 x 10 mm front speaker envelope below the volume controls while retaining the display, connectors and mechanical clearance. This is a placement reservation; the exact speaker and finished acoustic chamber still need qualification.

The previous enclosure is not approved for this board. Regenerate the case from the current outline, socket protrusion, screen connector height, microphone openings and SD-card travel before printing it.

What goes where

Position as seen from the front	Reference / access
Upper-front and lower-front acoustic openings	U20 / U21. Microphone chips mount on the rear and listen through 0.60 mm unplated PCB openings.
Three controls above the screen	SW1 Power; SW301 Start/Enter; SW302 Stop/Back.
Display and volume controls	45 x 31 mm off-board module guide. SW304 Volume- on the left; SW303 Volume+ on the right below it.
Left and right sides	J8 MIC A left; J9 MIC B right. Independent 3.5 mm analog inputs; socket bushings protrude beyond the PCB.
Right side and bottom edge	J6 microSD access on the right; J1 USB-C at the bottom for charging and native USB service.
Rear / internal harnesses	SW300 BOOT and SW305 RESET service controls. J2 battery; J3 NTC; J5 speaker; J7 display.

Rear views are mirrored: a front-right connector appears on the left when the board is turned over. PCB reference markings are the authority when checking harnesses.

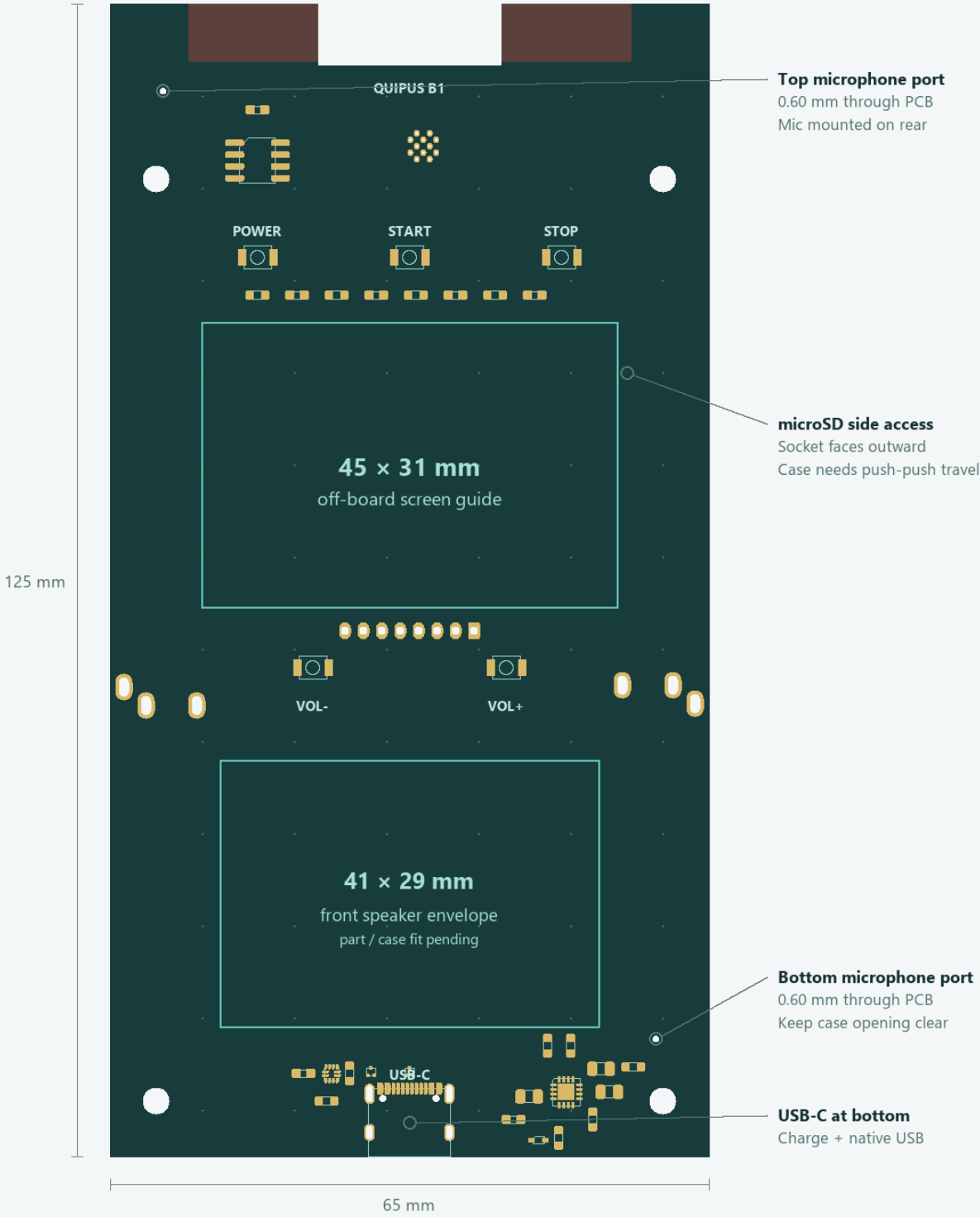
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Physical PCB placement • actual KiCad file geometry

65 × 125 mm outline | 1.6 mm board | 4 copper layers | 180 electronic references

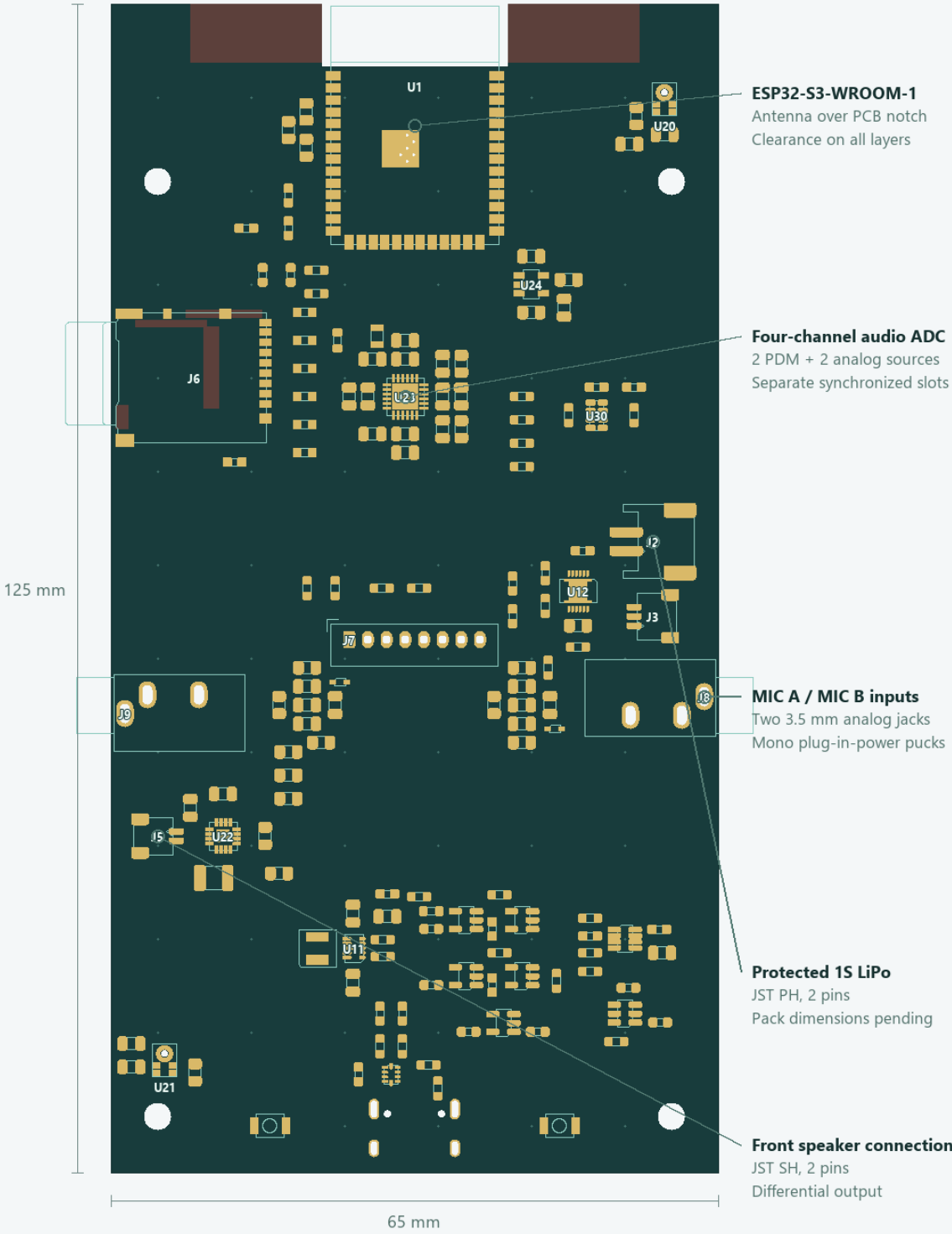
FRONT / user controls

Component-side view



REAR / electronics

Turn board over; left/right are mirrored



Physical copper pads Actual component outlines Copper keepouts Screen / speaker outlines are placement guides

No tracks or ground pours: the board is not routed. This drawing is derived from actual PCB geometry; it is not a rendered 3D model.

Source: Quipus-B1.kicad_pcb | SHA-256 d19ff1b295be70aafc89c9cb... | Rear mirrored; all geometry in millimetres

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Selected component overview

This is an orientation list. Purchase from the current full BOM only after its HOLD items are resolved.

References	Qty	Function	Selected part candidate
U1	1	Processor	ESP32-S3-WROOM-1-N16R8
U20 / U21	2	Built-in microphones	IM69D128SV01XTMA1
U23	1	Four-channel audio capture	TLV320ADC5140IRTWR
U24	1	PDM clock buffer	SN74LVC1G17DBVR
J8 / J9	2	Wired microphone inputs	SJ1-3533NG
U22	1	Speaker amplifier	MAX98357AETE+
U2 / U3	1 each	Charger / USB-C source detection	BQ24074RGTR / TUSB320LAIRWBR
U9 / U10	1 each	Power-button control / load switch	LTC2951ITS8-1#TRPBF / TPS22918DBVR
U11 / U12	1 each	3.3 V converter / battery gauge	TPS63802DLAR / BQ27441DRZR-G1A
U30 / U31	1 each	Offline clock / button expander	RV-3028-C7-TA-QC / PCA9536DR
J6	1	microSD socket	DM3AT-SF-PEJM5
SW1, SW300-SW305	7	Power, user and service buttons	B3U-1000P

Full parts list and off-board items

Full BOM: [output/pcb/Quipus-B1-EasyEDA/bill-of-materials.csv](#)

Includes all 180 references, electrical specifications, packages, assembly responsibilities, DNP decisions, purchase holds and source links. Keep the CSV with this addendum when requesting a quote.

Open accompanying [bill-of-materials.csv](#)

Off-board parts: Waveshare SKU 15867 screen module; approved protected 1S 2,000 mAh LiPo and cell-mounted NTC; qualified 8 ohm speaker; genuine microSD card; two PUI AOM-5024L-HD-R microphone capsules with shielded one-metre leads, matching TRS plugs and puck housings. Battery, NTC limits, speaker and case fit remain open selections.

Finish these before ordering an assembled board

1. Refine pin-side decoupling and converter current loops; complete power/ground, USB, audio and analog-input routing against a chosen stackup.
2. Qualify exact passive parts, ferrite, jack plated slots, microphone stencil and battery/NTC charging limits. Socket TVS protection needs circuit/transient verification.
3. Run native ERC/DRC and inspect copper, mask, paste and drill exports. Validate case and harness heights.
4. Bring up an assembled first article using current-limited power, then implement/test four-channel capture, SD retention, charging and complete uploads. USB microphone host support is not included.

Local checks report consistent pin/pad/net geometry and a packed placement. They do not certify native CAD checks, room pickup, audio latency, charging safety or battery endurance.